A negative-control ledger for multi-stage numerical pipelines, demonstrated on tensor-network benchmarks

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Abstract

We state an operational criterion for a genuine *negative control* in multi-stage numerical pipelines: **it must be able to fail when the conclusion does not hold**. Applying this criterion excludes positivity, finiteness, and non-emptiness checks, and sorts surviving controls into four types (repulsion, null-model, two-path, upper-bound). The ledger built from them carries a *completeness guard* that checks cited guards against the runtime registry, making the ledger hard to fake. Checks whose guarded proposition is measurably false are registered as *diagnostics* and excluded from the pass-rate denominator.

We demonstrate this method on a seven-layer pipeline whose engine is a transverse-field Ising / MERA tensor-network construction. **Results**: 57 of 57 scored assertions hold, 19 of 19 metrics meet target, and 13 negative controls cover all 7 layers. The substrate is pinned by a parameter-free CFT finite-size formula (relative residual 1.81×10^{-6} at L=16). A low-energy spectral ratio supplies a second estimator (0.212% deviation at $L \leq 16$), which is a *self-consistency check, not an assumption-free measurement*—the deviation grows monotonically at L=18/20 (0.284%/0.361%), falsifying a finite-size-correction interpretation.

We then turn the discipline on the ledger itself, reporting five self-audit outcomes: (i) coverage closed (6/7 → 7/7) but the layer was *not promoted*—the new control shares its construction with the criterion it guards; (ii) a registered "derivation" was a *tautology*; (iii) two registrations were *stale*, one concerning the denominator itself; (iv) a refuted inference was found in *four* places; (v) a guard too weak to fail printed a *plausible-looking fake reading*—the very failure mode the method exists to prevent.

Scope-negative-control extensions. We test whether the ledger is bound to critical CFT by re-running its gapped-state criterion on the spin-1 AKLT chain, whose exact $\chi = 2$ MPS ground state supplies an analytic oracle: the criterion returns the known area law ($c = 0$), and three constructive negative controls still fail as designed. Conversely, an audit of two *cited* tensor-network constructions finds source-level defects: the core matrix Y of the hyperinvariant MERA (hyMERA) appendix fails unitarity with residual $2|c_1 s_1|$ (6.742879×10^{-1}), while in the tensor-network renormalization (TNR) appendix Eq. (TRG2) is non-positive-definite (minimum eigenvalue -2.0) and mutually exclusive with Eq. (TRG4). These are defects of the cited literature, not of our implementation.

Boundary. This is a *methodological* report: it establishes the feasibility and limitations

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of a self-auditing pipeline workflow, not the validity of MERA as a description of physical systems. The tensor-network pipeline is a computational testbed; we validate the auditing discipline on it, not the other way around. We claim no physical result beyond textbook Ising criticality, and "100%" means only that 57 scored assertions hold. The maturity vector is A × 1, B × 3, C × 3, D × 0, and no layer was promoted.

Code availability: `https://github.com/yanjianzhong/spiral-emergence`

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1. Introduction

1.1. The problem: a sanity check is not a negative control

A layered pipeline derives quantities stage by stage. At each stage an assertion is made — "the spectrum is degenerate", "the entropy is positive", "the grid is fine enough" — and downstream stages consume the derived quantity. In practice these assertions are enforced by *positivity*, *finiteness*, and *non-emptiness* checks, often promoted from a hard `raise` into a readable boolean.

The difficulty is structural: such checks are *consistent with any conclusion.* "Schmidt rank ≥ 2" holds for a correct calculation and for a calculation wrong in every way that matters [1]. A pipeline can therefore report a wall of green checks while every substantive claim in it is false.

This paper is about a discipline that fixes that, and about what happens when it is applied to our own code and our own recorded inferences.

1.2. Contributions

1. **A criterion** (Sec. 2.1) separating genuine negative controls from true-by-construction checks, explicitly *excluding* positivity/finiteness/non-emptiness.
2. **A taxonomy** (Sec. 2.2) of four control types, each tied to a concrete implemented instance.
3. **A ledger with a completeness guard** (Sec. 2.3) that checks against the *runtime* registry rather than trusting the ledger's self-report.
4. **A runtime partition** (Sec. 2.4) replacing two hand-maintained lists with a single registry, so the denominator cannot drift.
5. **A demonstration** (Sec. 3–4) on a seven-layer pipeline, including three negative results.
6. **Five self-audit outcomes** (Sec. 6), each generalising to a rule.
7. **Two scope tests** (Sec. 9 and 10): the ledger on a non-critical benchmark, and an audit of two cited tensor-network constructions.
8. **Per-layer honest boundaries** (Sec. 11), including v16.4 assessment of all seven layers.

1.3. Scope, and what this paper does not claim

Stated early, because it governs how every number below should be read:

- We do **not** claim the seven-layer framework is established (maturity A × 1, B × 3, C × 3, D × 0; Sec. 11).

- We do **not** claim a holographic bridge was established between layers 4 and 5 (Sec. 5.3); tensor-network/holographic correspondence is only background motivation [2].
- 100% pass rate **does not** mean the evidence became stronger (Sec. 12.2).
- The substrate physics — critical Ising/CFT — is **textbook material**, verified as a substrate, not offered as a result.
- We do **not** claim MERA cannot produce state-dependent geometry in general; only that in the present fixed-wiring binary implementation the readout is an analytic property of the graph layout (Sec. 5.4).
- We do **not** claim that this work validates MERA as a physical model; the pipeline serves as a *testbed for the auditing methodology*, not as evidence for tensor-network descriptions of holography or emergent geometry.
- Layer 7 is a philosophical/conceptual endpoint, not in competition with existing scientific theories (Sec. 7).
- The AKLT results (Sec. 9) test one criterion and three negative controls; they do **not** establish that the ledger is generically applicable to non-critical systems.
- The hyMERA and TNR findings (Sec. 10) concern source-level defects in *cited* literature; they are not claims about our own implementation.

The contribution is methodological: the novelty lives in the *method layer* — the claim that the validation framework is itself subject to the same logical standard it imposes on the pipeline.

2. Method: criterion, taxonomy, and ledger

2.1. The criterion

> **Criterion.** A check qualifies as a negative control only if there exists a concrete, reachable state of the pipeline in which *the conclusion it guards is false* and the check *fails*.

The criterion is applied as a filter. In this pipeline it removed an existing candidate — "the entanglement entropy is positive" at layer 4. Positivity holds for a wrong entropy as readily as for a right one; it is now labelled a sanity check. The same filter excludes the whole positivity/finiteness/non-emptiness family. This exclusion is recorded in the control-registry header in the source (`_NEG_CTRL_TABLE`), so it is a property of the code rather than of this prose.

2.2. Four types of control

Every control admitted by the criterion is assigned one of four types. Each type is defined together with the instance that actually implements it in the pipeline:

2.3. The ledger and its completeness guard

The ledger groups controls by physical layer, and each entry cites the guard name(s) that implement it. A ledger is a self-report, and a self-report can be written as a brochure: it can cite guards that do not exist, or quietly omit a layer.

The completeness guard closes that hole by checking the ledger against the *runtime registry* of guards rather than against its own text:

Type	Definition	Implemented instance
Repulsion	A quantity is pushed away from a value it would take if the conclusion were false; the observed displacement is the evidence.	Layer 4 bond dimension: when the bond dimension is forced below its lower bound, the downstream quantity degrades by 6.47× (criterion > 1.5×). L1 extension: rotate the local state and observe overlap drop below 0.9 (Sec. 7.3).
Null model	The measurement is compared against an ensemble generated to share the *nuisance* structure of the real object while destroying the structure under test.	Layer 5 bulk geometry: three null families × 3 instances — `gnm` (matched $\|V\|$, $\|E\|$; controls scale), `ws` (degree-matched small world; controls degree distribution), `tree` (balanced binary tree; controls the "nearly a tree" premise). Two-sided acceptance band $[5\%, 95\%]$.
Two-path	The same quantity is obtained by a second, structurally different route; agreement is the evidence.	Five instances, spread across four layers (L2 ×2, L3, L5, L6) — for example the L2 spectral-gap degeneracy guard.
Upper bound	A computed quantity must not exceed an independently known bound.	Layer 7: a single elementary bound (the thinnest control in the ledger). L1 extension: the equal-state overlap against the exact ground state.

Table 1: The four control types, with the instance that implements each. Type counts in the current ledger: repulsion 6, two-path 5, null-model 1, upper-bound 1 (13 total).

- every guard name cited by the ledger must exist in the live registry;
- each of the seven layers must appear exactly once.

In the reference run this guard reports `phantom references: none; missing layers: none`. It caught a defect in which a completeness criterion was inverted (Sec. 4.4).

> **What this guard is, and is not.** It makes the *method itself* falsifiable: a ledger that cites phantom guards or silently omits layers is, in effect, a falsified document. It is not a physical conclusion — it guarantees only that the ledger contains no phantom citation and no silently dropped layer. Its object is the honesty of the accounting, not the truth of the physics. The defect caught in Sec. 4.4 is proof that this self-falsifiability is not a slogan: it caught a bug *in the validation layer*, exactly where bugs are hardest to see.

2.4. Scoring items versus diagnostics

Some guards protect propositions that are *measurably false* in the system as built, or that depend on a parameter rather than on the system. Reporting these as failures would be misleading, and silently dropping them would be worse. They are therefore registered as *diagnostics* and excluded from the pass-rate denominator.

In the reference run: 78 guards total, of which 57 are scored and 21 are diagnostics. **Of the 21 diagnostics, 16 fail by design.** The `expect_pass` flag is set explicitly at

registration, so the distinction is a property of the code and survives any re-reading of the output.

Runtime partition.. Earlier iterations kept scored checks and diagnostics in two separate lists with a hard-coded denominator — fragile because a mismatch is silent. The current implementation uses a single registry with partition computed at report time:

> `guard_pass_rate()` iterates the *same* `GUARDS` list and splits it by `expect_pass`: `True` → scored; remainder → diagnostics. `v16/_metric/registry.py:153-161`

The denominator is therefore derived from the same object the checks were recorded into — not a number anyone can transcribe incorrectly.

Why this matters. Admitting checks that cannot fail inflates the score. Separating the denominator costs one boolean per guard.

3. The demonstration pipeline

3.1. Seven layers

The pipeline derives, layer by layer, from an exact-diagonalisation (ED) solution of the critical transverse-field Ising chain through entanglement spectra, bond dimensions, tensor-network bulk geometry, and a downstream scaling analysis. Each layer exposes (i) the proposition it asserts, (ii) the derived quantity it hands downstream, and (iii) its downstream consumers. The audit table records, per layer, which downstream consumer actually reads the derived quantity — this is what makes the gap in Sec. 5.2 visible.

3.2. Substrate and its constraint

The validation substrate is the critical transverse-field Ising chain: $c = 1/2$, $v = 2$, $e_\infty = -4/\pi$. A second, non-critical tensor-network benchmark — the spin-1 AKLT chain with an exact $\chi = 2$ MPS ground state — is introduced in Sec. 9 specifically to test the scope of the ledger.

One boundary deserves emphasis because it is easy to misattribute. The pipeline's chain-length ceiling ($L \leq 16$ on the closed loop) comes from a **tensor-network library constraint**: the MERA construction used [3] requires L to be a power of two. It is *not* an exact-diagonalisation limit (sparse ED reaches $L = 20$ in 22.4 s) and *not* a laptop budget. This cause was mis-stated twice in the source comments before being corrected; it is reported here because the same misattribution is easy to make elsewhere.

4. Results from the initial demonstration

4.1. Coverage of the ledger, and its gap

The initial ledger (v15) had 12 negative controls covering **6 of the 7 layers**; the gap was `['L1']`. The strongest layer was L5 (three null families × 3 instances, two-sided band); the thinnest was L7, with a single upper-bound control. The per-layer coverage guard was *deliberately not scored*: coverage is a to-do list, not a property of the model, and scoring it would be grading the checklist.

Quantity	Reference run (v16.3)	Note
Scored assertions	$57/57 = 100.00\%$	see Sec. 5.1
Diagnostics excluded	21 (16 failing by design)	`expect_pass=False`
Metrics meeting target	19/19	27 registered; 8 diagnostics excluded
Negative controls	13	repulsion 6 / two-path 5 / null 1 / bound 1
Layer coverage	7/7, gap `[]`	gap was `['L1']` in v15
Ledger completeness	no phantom refs, no missing layer	Sec. 2.3
Wall-clock (8 cores)	1119.7 s	49.9% aggregate CPU utilisation

Table 2: Reference-run summary. Values transcribed from the pipeline log of the frozen reference run `v16/_runall_v16.3_r1.log`. The metric denominator excludes the one layer that carries no target line — see Sec. 12.2.

4.2. External anchor: the $1/L^2$ coefficient, at every L

The substrate is pinned by a comparison to a formula with *no free parameters* [4]:

$$\frac{E_0(L)}{L} \;=\; e_\infty \;-\; \frac{\pi c v}{6L^2}, \qquad c = \tfrac{1}{2}, \quad v = 2, \quad e_\infty = -\tfrac{4}{\pi}. \tag{1}$$

E_0 comes from exact diagonalisation; the constants on the right do not enter its computation.

L	relative residual vs. Eq. (1)	residual-inverted power p
8	2.88×10^{-5}	4.007
⋮	⋮	⋮
16	$\mathbf{1.81 \times 10^{-6}}$	4.004
18	1.13×10^{-6}	4.002

Table 3: The residual decreases monotonically with L, and the residual-inverted power is $p = 4.00$ throughout — the expected leading correction. The anchor script carries a `--selftest` that demonstrates the comparison is not identically true.

Honest boundary. This is stronger than the anchor already present in the code, which compared only the *extrapolated* e_∞ against $-4/\pi$ (deviation 5.0×10^{-6}); here the comparison is against the *per-L* $1/L^2$ coefficient, which involves the product cv. But it remains a statement that *exact diagonalisation was performed correctly.* It is textbook content, not a result of this paper.

4.3. The central charge: self-consistency, not assumption-free measurement

A second estimator for the central charge uses the low-energy spectral ratio $(E_2-E_0)/(E_1-E_0)$, which is independent of v:

Two hard boundaries, both required reading. *(i)* The estimator is a self-consistency check, not an assumption-free measurement: the operator content is identified before use (E_1 is the R-sector vacuum, $\Delta_1 = 1/8$; E_2 is the NS two-particle state, $\Delta_2 = 1$), so $\Delta_1 = 1/8$ enters the construction. *(ii)* The readout does **not** converge monotonically in L, and the 0.212% figure is valid only for $L \le 16$. At L=18 the deviation is 0.284% and at L=20 it is 0.361% — $|c - 0.5|$ grows *monotonically* from L=10 (5.3×10^{-4}) to L=20 (1.806×10^{-3}), a

Quantity	Value	Note
ratio at L=8	7.9231	monotone toward 8
ratio at L=16	7.9807	best in $L \leq 16$
ratio at L=18	7.9758	deviation 0.284%
ratio at L=20	7.9699	deviation 0.361%
c interval ($L \leq 16$)	[0.49894, 0.50034]	max deviation 0.212%

Table 4: Low-energy spectral estimator. Both the entanglement-spectrum fit and the spectral ratio contain c as a prior; they are consistency checks, not assumption-free measurements.

27.8× increase that never reverses. The earlier $O(1/L)$ finite-size-correction interpretation is therefore **falsified**: $O(1/L)$ would imply decay, not growth. The correct statement is that the ratio *drifts away from 8* from L=10 onward; it is a self-consistency check on the $L \leq 16$ window, not an extrapolation to $c = 0.5$. The 0.212% figure and the "ratio → 8" statement are *the same measurement*: 0.212% *is* the deviation by which the ratio misses 8. It must not be written as "the spectral family and the Rényi family corroborate each other"; the correct statement is that the spectral family is *independent of the entanglement-spectrum family.*

4.4. Negative controls caught a real defect

The strongest argument that the controls are not decorative is that one of them caught a defect in which **a completeness criterion was inverted**.

At three places the ledger wrote "the XX negative control has been run" — a *self-reported* name — into the same tuple as the actually-executed guard names. Such names are registered only on an early-return branch, and therefore *never enter the registry when the control executes normally.* The consequence was exactly backwards: the completeness check **failed in the healthy case and passed when the control was missing**. Phantom references: 3 before the fix, none after.

What makes this worth reporting. The defect was found only by a *full pipeline run.* Before that, a source-text-versus-regex audit had reported 26/26 correct. A source-text audit is necessary but not sufficient: it cannot see runtime registration paths. The defect was introduced by the present work, not inherited; it is reported because the failure mode — a check that passes for the wrong reason — is the exact failure this entire method exists to prevent.

5. Honest boundaries from the initial demonstration

5.1. Twenty-one diagnostics, sixteen failing by design

Log line, quoted: `guards 57/57 = 100.00% (plus 21 diagnostics excluded, of which 16 did not pass).`

"Did not pass" is not a synonym for "defective". These 16 are *archived negative results.* Key examples:

- **A third corroborating path does not exist.**
- **The MERA bulk readout does not depend on the state** (Sec. 5.4).
- **That layer has no causal path** (Sec. 5.2).

- **The spectral radius is parameter-dependent**, not a property of the system.

By contrast, the one addition that *does* enter the denominator can fail: if the bond-dimension derivation were replaced by a hand-picked constant, its control would flip. The threshold there ($> 1.5\times$) is an order-of-magnitude comparison, and the line it references is not a strict lower bound — both facts are recorded in the guard's own note.

5.2. Gap 1 — a layer where no control can be constructed

For layer L1 the ledger initially reported no negative control. The cause has been traced, and it is not "nobody got round to it".

The quantity L1 hands downstream, `equal_state`, is **never consumed anywhere in the repository**: it appears exactly once, in its own `return` statement. Sibling quantities are no better — a condition number is reported but nothing is checked against it; a dimension is fed into a downstream derivation whose result is *printed* but never compared against $2^{L_{\text{chain}}}$.

There is no quantity to push away from, because there is no downstream consumer to be affected. The gap is registered as a *source-level attribute-access scan*, which is falsifiable in the useful direction: should anyone connect the quantity, the scan flips. A smoke test injects a fake consumer to demonstrate that the scan is not identically false.

This gap is "empty for a reason", not "left unfilled". It has deliberately *not* been entered into the L1 row of the ledger — doing so would pretend that L1 has a negative control. This gap was closed in v16 (Sec. 6.1), but the layer was *not* promoted.

5.3. Gap 2 — a bridge that is wired, and still carries nothing

An earlier internal note claimed that the L4→L5 bridge was *empty*. Direct reading of the call sites shows that claim was wrong: the bridge is wired end to end — the MERA graph is built, the curvature report is computed from it, the forman value is derived from it, and the control family is constructed on it.

The controls are not missing either; they are the strongest in the ledger (three null families $\times$ 3 instances, two-sided band). Measured outcomes:

- **F6b** (the MERA bulk geometry lies within the control distribution): *passes*. Forman percentile 33%, Ollivier percentile 33% inside the $[5\%, 95\%]$ band.
- **F6c** (the MERA geometry is more negatively curved than *all* controls): *false, by design excluded*. Per family, Forman: `gnm` -3.553, `tree` -0.867, `ws` -0.194, versus MERA -2.242. The `gnm` control is *more* negative than the MERA graph — the escalation claim does not even have the right sign.
- **F6d** (the readout depends on the state, or on the bond dimension): *false, by design excluded*. See Sec. 5.4.

So the gap is neither "not wired" nor "no control available". It is a third kind: **no upstream-derived quantity flows along the wire.** The quantity L4 derives ($\chi_{\text{dev}} = 4$, from the area law) is consumed elsewhere — its consumers were enumerated by search — and *none of them leads to the L5 construction*, which hard-codes its own bond dimension.

5.4. The "geometry" readout is an analytic property of the wiring

Layer 5's bulk-geometry readout is computed on a MERA tensor-network graph. Within the present implementation—a fixed-wiring binary MERA where the graph builder reads only the index map—the readout does not depend on the fitted state, seed, or bond dimension (Table 5). Three measurements show this:

Varied	**Method**	**Result**
seed $(0, 1, 7)$	rebuild and compare edge sets	**identical**, bit for bit
bond dimension χ $(2, 4, 8)$	rebuild and compare edge sets	**identical**, bit for bit
state	the graph builder reads only the index map; the fit routines only replace tensor data	wiring never changes

Table 5: The MERA bulk graph does not depend on the state, on the seed, or on the bond dimension in the current fixed-wiring implementation.

Honest boundary. In homogeneous MERA tensor networks, many structural properties are predetermined by the network geometry and are largely independent of variational parameters [5]. This is a normal property of the architecture; what it means here is that the readout is a property of the **binary-layout wiring**, and the upstream claim must be re-worded accordingly. It is therefore *not* a state-dependent emergent geometry under the present implementation. Whether a modified architecture (e.g., cMERA, state-dependent disentanglers, or gauged/SymTFT lift) restores state dependence is left to future work following Swingle's entanglement-renormalization approach [2]. Reported values: $|V| = 62$, $|E| = 91$, $\bar{d} = 2.935$, **triangles** $= 0$, Forman mean -2.242, Ollivier mean -0.3297, percentiles 33%/33%.

5.5. Two self-contradictory conventions, corrected

Two reporting conventions internal to the project contradicted each other and were corrected rather than reconciled by wording: (i) a claim that a scatter statistic was "inter-seed spread on the reference side" — contradicted by the archived fields; (ii) a claim that two routes "corroborate each other" — contradicted by the fact that they are the same measurement (Sec. 4.3). Both are recorded as corrections, with the original documentation left unedited and the discrepancy noted.

6. Turning the method on itself: five self-audit outcomes

This section reports what happened when the same discipline was turned on the ledger's own author and code. The result is not a new physical claim. It is a catalogue of the ways a verification discipline *actually* fails in practice, each instance observed rather than hypothesised — and, most usefully, an instance in which the failure was caught by the discipline itself rather than by an external reviewer. We take that last class to be the strongest available evidence that a verification method is doing work: a method that has never caught its own author is indistinguishable from a method that cannot.

	what was caught	disposition
§6.1 gap closure	coverage went $6/7 \to 7/7$, but the new control shares construction with the criterion it guards	layer **not** promoted; the control calibrates *sensitivity*, not independent falsification
§6.2 tautology	a registered "derivation" was two juxtaposed theorems; the entropy object in the claim was not the entropy the code computes	registered as a *tautology*; the gap it was meant to close is **not** narrowed
§6.3 stale registration	two registry entries described behaviour the code does not have; one touched the pass-rate denominator itself	corrected in place, original retained and annotated; denominator settled from code, not from a log line
§6.4 refuted inference	one inference had been written in **four** places before being refuted	swept repo-wide by equal-length replacement, not patched at the site of discovery
§6.5 a guard that cannot fail	a guard passed while its guarded quantity was meaningless, printing a plausible fake reading	guard strengthened to key on the maximum entry; the incident is registered as a live instance of the paper's own thesis

Table 6: The five self-audit outcomes. None of them changes a physical readout; all of them change what the ledger is entitled to say.

6.1. Closing a gap is not the same currency as adding evidence

The initial ledger had a single uncovered layer, $L1$: it had no negative control. The v16 iteration closed that gap — coverage is now $7/7$, gap `[]`, with 13 controls.

The natural next sentence is "therefore $L1$ is promoted". We did not write it. The reason is visible only by reading the control's implementation: the repulsion control evaluates $\left|\langle R_y(\theta)\psi^{\otimes L} \mid g_{\mathrm{ED}}\rangle\right|^2$, where the trial state is prepared by *the same* `np.kron` construction that the primary criterion uses. A control built from the criterion's own construction cannot falsify that criterion; it can only calibrate the scale on which the criterion's sensitivity is measured. That property is real and worth having — and it is recorded in the control's own note — but it is not independent evidence.

Furthermore, the overlap calibrates a *sensitivity scale*, not an independent physical falsification. And the statement "`equal_state` is the $h \to \infty$ limit" is a *mathematical fact*; calling it "the L1→L2 causal connection" is a *terminology choice* — these two sentences must not be conflated (since h is finite, $\mathrm{overlap}_\infty = 0.999024 < 1$).

The layer therefore remains at its previous maturity, and the ledger records the promotion as *considered and declined*, with reasons.

> **Rule.** "The gap is closed" and "the evidence is now independent" are different currencies. A coverage counter cannot tell them apart, because both increment it. Only reading the control's construction can.

6.2. The method caught a derivation that was a tautology

One registered item claimed to derive a uniqueness statement about a measure. Auditing it against the code showed the "derivation" to be two ready-made theorems placed side by side: that the uniform distribution maximises Shannon entropy, and that the Haar measure is the unique unitary-invariant measure. Juxtaposition is not derivation, and neither theorem constrains the other.

The audit also found a mismatch in the *object*: the claim's entropy is the Shannon entropy of a probability vector, whereas the code's state is a pure state, whose von Neumann entropy is identically zero — a *minimum*, not a maximum, of that functional. The two are not the same quantity.

Disposition: registered as a tautology, with the explicit statement that a tautology **does not narrow the gap** it was invoked to close. This is the same disposition the initial demonstration gave its own reductive results, now applied reflexively.

> **Rule.** A "derivation" that only re-labels structure already assumed contributes zero. Audit the derivation's object, not just its conclusion.

6.3. The method caught two stale registrations, one of them about the denominator

Two registry entries were found to describe code behaviour that does not exist.

(a) A verdict recorded backwards.. A ledger comment stated that two checks, $V6$ and $V7$, are both registered with `expect_pass=False` and therefore excluded from the pass rate. Reading the call sites shows that only $V6$ carries that flag; $V7$ is scored. The registry had been describing a *verdict* that the code does not implement. This is not cosmetic: had the comment been taken as authoritative and "corrected" in the other direction — by adding the flag to $V7$ — the scored denominator would have dropped from 57 to 56.

(b) A guard count settled by code, not by a log line.. Two documents and one script banner stated the guard total as 77. The frozen run's report line gives 57 scored $+\,21$ diagnostics $= 78$, and the partition function splits a single list — so 78 is a consequence of the current code, whereas 77 was a consequence of the code as it stood at an earlier date.

This is the point at which the discipline acquired a rule about *tense*. An audit found "77" in eight places. Four of them are *dated archive records* — entries explicitly labelled with an earlier version and date, in which 77 was the number genuinely measured at the time. Those four were left *unchanged*. Editing them to 78 would not be a correction; it would be falsification of the record. The other four were standing assertions about the current value and were corrected.

> **Rule.** The same digit may be stale or true depending on tense. A dated archive assertion is never "updated"; a present-tense assertion is never left stale. Before changing a count, broaden the search: the registered list of sites may itself be miscategorised, and correcting only the listed subset leaves two mutually contradictory documents.

6.4. The method caught a refuted inference in four places

An earlier inference — that a particular measurement *established* that a certain carrier dimension is not excluded by a no-go argument — was refuted when the measurement was recomputed. The refutation was local, but the inference was not: it had been written into four separate documents, including a runner's explanatory note and a script's header.

Disposition: the four sites were corrected in one batch, by *equal-length replacement* — substituting N lines for N lines — because one of the affected files is cited by line number from a smoke test, and inserting a line would silently break that citation.

> **Rule.** An inference refuted at one site is not refuted; it is refuted only when every site carrying it is swept. And when the swept files are cited by line number, the sweep must be length-preserving, or it will manufacture a second class of error while fixing the first.

That second class duly appeared: a mid-file insertion into a document moved the line numbers of citations *into that same document*, and a section inserted at the *top* of a changelog moved a set of line-number citations by roughly 150 lines. The content was correct throughout; only the pointers drifted. The disposition adopted is to convert fragile line-number citations into verbatim-fragment citations, and to accept that the already-drifted pointers are registered rather than chased — since editing the body to realign them would move other lines, and recurse.

6.5. A guard that could not fail printed a plausible fake reading

This is the paper's centrepiece, and it happened inside the project's own reference implementation, while it was building a new downstream readout for $L1$.

The setting.. The readout propagates a trial state with the imaginary-time operator $e^{-\tau H}$ — H the periodic transverse-field Ising Hamiltonian at $L{=}16$ — and measures the half-chain entropy $S(L/2)$ downstream. The natural time scale is $\tau^* = 1/|E_0/L|$; with the registered $E_0/L = -1.275287$, $\tau^* = 0.784137$.

The trap.. $e^{-\tau H}$ is *upward* propagation in imaginary time: the norm grows as $e^{\tau|E_0|}$. With $|E_0| = 20.4046$ at $L{=}16$, the growth factor at $\tau{=}20$ is e^{408}, and the vector's largest entry is around 2×10^{176} — comfortably representable. The implementation's guard was

```
if not np.all(np.isfinite(v)):  # report unavailable
```

which *passes*, since every entry is finite. Normalisation was then `u = v / np.linalg.norm(v)`. But `np.linalg.norm` squares before it sums; with entries of order 10^{176} the square overflows to `inf`, the norm is `inf`, and `v / inf` is the *zero vector*. The subsequent entropy and overlap are computed from that zero vector and printed:

$$S(L/2) = 0, \qquad \big|\langle g_{\mathrm{gs}} \mid u\rangle\big| = 0.$$

Both are finite, well-typed, and formatted like every other reading in the report. Nothing in the output distinguishes this from a genuine result.

The fix.. Guard on the *maximum entry* rather than the norm, and scale by it before normalising — a quantity that has not yet been squared, and so cannot overflow while the entries themselves remain representable. The overflow bound then becomes a readable fact rather than a crash: $\tau > 709.78/|E_0| = 34.79$.

Why this belongs in a methodology paper.. The thesis of this work is that a sanity check is consistent with any conclusion, so a wall of green checks does not distinguish a correct pipeline from a broken one. Here the check in question — finiteness — is precisely a member of the excluded family, it was written by an author who had stated the exclusion criterion, and it *passed* while its guarded quantity was meaningless. The output was not an error message; it was a number.

The episode is registered as three numerical facts, verbatim from the run's own transcript: at τ=10, $\max|v| = 5.063 \times 10^{87}$; at τ=20, 2.092×10^{176}; at τ=30, 8.641×10^{264} — each with the guarded readings that replaced the fake one. We note that the same episode supplies a second, smaller instance of Sec. 6.3's rule: an author comment stating the tolerance rationale was itself initially wrong (it quoted a half-last-digit derived from a four-decimal printed value rather than the three-decimal value actually printed), and was corrected before the run was registered.

> **Rule.** A guard must be keyed on a quantity that *cannot* remain finite while the guarded value becomes meaningless. "All entries are finite" is not such a quantity when the next operation is a sum of squares.

6.6. What the five share

Each of the five is an instance of the same structure: *a check that incremented correctly while the thing it stood for did not obtain.*

- Coverage incremented while independence did not.
- A derivation was present while a derivation did not occur.
- A registry entry existed while its described behaviour did not.
- A correction was applied while the refuted claim survived elsewhere.
- A finiteness check passed while the value was meaningless.

The composite lesson is that the failure modes are *bookkeeping-shaped*, not arithmetic-shaped. None of the five would be caught by a numerical tolerance, a longer run, or a larger L. All five were caught by asking what the check's increment *means*.

7. Negative controls: which ones can actually fail

A control that has never been observed to fail is indistinguishable from a control that cannot. This section tests the controls by scanning them over declared grids.

7.1. Four orders of magnitude between a check and its downstream

For the $L4$ layer, an isometry-destroying perturbation of strength ε was injected along an isolated path (deliberately bypassing the main entry point, whose registered readout must not move). Result: a perturbation of $\varepsilon = 10^{-6}$ already drives the isometry residual above

10^{-12}, whereas the downstream central-charge readout does not degrade until $\varepsilon \approx 3 \times 10^{-2}$ — a gap of about four orders of magnitude. Two further properties are registered alongside, because reporting only the first would overstate the control: the response is *non-monotone* (at $\varepsilon = 10^{-3}$ the downstream figure is $0.787\times$ its baseline, i.e. *below* it), and the absolute value of the extracted central charge is not comparable to production, since the perturbed path runs at a different step count. It is the *gap* that is the finding, not the endpoint value.

> **Rule.** A sensitivity margin is a quantitative result and should be reported as one — together with any non-monotonicity, which a single "it fails" summary would hide. The margin must be reported on the same page as the control, or the control will be read as stronger than it is.

7.2. Where the discriminating power actually comes from

For the $L6$ reaction–diffusion layer, the control's two free parameters were scanned over a 30-point grid, frozen *before* the scan, with the freeze line recorded in the log. On that grid the criterion reports "not emerged" at 23 of the 30 points, and all 30 are flagged finite — so the criterion can fail, and its failure is not an artefact of a numerical breakdown.

Re-partitioning the already-recorded table (no new data, no point selection) localises the power further: all 23 failing points have a contrast of exactly zero, whereas only 11 of them also satisfy the activity bound — the other 12 are uniformity-saturated states. On this grid, discrimination comes from the contrast criterion *alone*. We state this for this declared grid only.

Boundary: The dual-path convergence (10^{-14}–10^{-17}) is a *code-fidelity* check — it verifies the equations are solved correctly, not that the equations describe life. The Gray–Scott system produces Turing patterns; this is a standard reaction–diffusion result, not a model of life origin. Key gradient parameters remain hand-set, not derived from upstream layers. Elevating this to a scientific claim about life would require a dissipative-adaptation or chemical-Darwinism framework, which is outside this work's scope.

7.3. A downstream path that exists, and can fail, and is still not evidence

The $L1$ layer admits a directly relevant test: propagate the layer's own state with $e^{-\tau H}$ and ask whether the downstream readout can distinguish it. It can. At τ^*, the correct state gives $S(L/2) = 0.5090065518$ while a deliberately wrong state gives 0.7150567405, a relative separation of 2.88×10^{-1} against a pre-registered threshold of 10^{-3}. The four registered readouts of the reference run are reproduced at relative deviations of 1.6×10^{-7}, 4.5×10^{-5}, 3.9×10^{-4} and 4.0×10^{-7}, and the entropy readout agrees with the pipeline's own reference value to 3.3×10^{-16}. The acceptance criterion — "it must be possible to fail at least once" — is met.

It is still not evidence for the layer, for three reasons, two of which were written down *before* the measurement:

1. The operator acts on a *trial state*; the production readout is the *ground state* of H. The path is a parallel construction, not the production readout, and its agreement therefore does not transfer.
2. "Not sharing a construction" holds only in how the state is written. The tensor is assembled by pointwise multiplication and reshaping rather than by `np.kron` recursion — verified

by counting `kron` attribute accesses in the syntax tree, giving zero — but the two are *numerically identical*. What is genuinely not shared is the readout mechanism, not the state preparation.

3. The layer's promotion conditions are conjunctive, and this addresses only one of them.

> **Rule.** "The check can fail" and "the check tests the right thing" are independent properties. Passing the first while missing the second is the normal case, not the exceptional one.

8. The one positive template: a criterion that could have chosen otherwise

Five of the self-audit outcomes above are negative. It is worth recording that the project contains exactly one result of the opposite kind, because it identifies what the negative ones lack.

In the cMERA branch ($L5$), a self-check of the Gaussian representation passes 9/9. What distinguishes it is not the pass rate but a structural property of how it was built: *the criterion selected the convention, rather than the convention determining the criterion.* The check was fixed first, and the representation convention was then the thing that had to satisfy it — so a different convention would have failed, which is exactly what the other five results cannot say about themselves.

Two boundaries must be attached, and they are why this section is short.

(i) It is a statement about the criterion.. It says that a convention was constrained, not that a spacetime emerged.

(ii) What it does not support.. A companion probe found that under the declared convention the cMERA flow's terminal state is *indistinguishable* from the bare ultraviolet vacuum, with a relative span of 2.285×10^{-15} across the objects compared. That is a negative result about the branch, and it is why $L5$'s maturity is unchanged. A separate controlled recalculation (same parameters, same units, N=36/72/144) eliminated a 252× unit-confounding artefact but placed the model side inside the reference side's own initial-value spread — a null result. The L4→L5 bridge therefore remains empty; the 6-dimensional structural comparison claim reduces to 4 effective dimensions (two descriptors degenerate to constants).

> **The constructive rule.** A result is a result, rather than a tautology, exactly when the criterion could have come out otherwise. This is the property whose absence the self-audit outcomes of Sec. 6 detect.

9. The AKLT chain: a non-critical scope-negative-control

9.1. Motivation: is the ledger bound to critical CFT?

All 13 negative controls of the main ledger run on the critical transverse-field Ising chain or objects derived from it. This leaves an unanswered criticism: *is this method valid only for CFT substrates?* The question is legitimate and, until now, unanswered by the ledger itself — a gap, not an answered problem.

The spin-1 AKLT chain supplies a decisive counter-instance. It is gapped (Haldane gap), satisfies an area law, has no CFT correspondence, and its ground state is the *exact* $\chi = 2$

MPS [6, 7] — no variational optimisation required. Every layer of the ledger therefore has an analytic ruler: a result known in advance against which to compare. This is precisely what a negative control requires — a case in which the conclusion *should* fail.

Scope discipline. This extension is a *scope-negative-control* for the method itself. It does **not** add a physical stage, does not promote any layer, changes no registered reading, and moves no denominator: the guard count remains 78, the metric count 27, and the negative-control count 13. It is recorded in an independent audit section (`spiral_v16_audit.md`, §9.15), not in the runner, and introduces no new guard or metric.

9.2. Results on the AKLT chain

The gapped-state criterion of the repository, `gapped:` $|c_{\texttt{scaling}}| < 0.05$, is fed the exact AKLT MPS. Results, for both open and periodic orientations, are given in Table 7.

Test	**Observed**	**Interpretation**
Gapped criterion $\lvert c_{\text{scaling}}\rvert < 0.05$	open: $c = -0.0000000$; $S(L/2) \equiv \ln 2$ (exact, L-independent, area law). periodic: $c = +0.0036173$	both pass; the known answer is $c = 0$
Bond-dimension reduction ($\chi : 2 \to 1$, product state)	$S(L/2) = -0.000\text{e}{+00}$ (genuine product state, Schmidt spectrum single-branch)	negative control **fails**, as designed
Non-unitary injection $B[1] \mathrel{+}= 0.05\,\lvert\downarrow\rangle\langle\uparrow\rvert$	isometry residual 2.886751×10^{-2} ($= 0.05/\sqrt{3}$); $\sum\langle P_2\rangle$ rises from $\sim 10^{-16}$ to 10^{-3}–10^{-2}	negative control fails; **six orders of magnitude** of bite
Isometry bidirectional pair	positive control $0.000\text{e}{+00}$ **and** negative control 2.887×10^{-2}	checker is neither rubber stamp nor blanket failure

Table 7: AKLT-chain tests. The gapped criterion returns the known area law $c = 0$; three constructive negative controls still fail as designed. The checker is thus discriminating in both directions.

The exact oracle family is verified to machine precision throughout: $\langle H\rangle = -\frac{2}{3}n_{\text{bonds}}$ (not zero — the trap is that the projector sum $\sum_{\text{bonds}} P_2$ vanishes, which is where the implementation-phase error occurred); open-chain midpoint $S = \ln 2$; periodic n=1 spectrum $(1/3, 1/3, 1/3) \Rightarrow S = \ln 3$; periodic-block spectrum showing SU(2) $1 \oplus 3$ degeneracy; and exact isometry 0.0 (subject to the global prefactor $2/\sqrt{3}$, without which $\sum_s A^{s\dagger}A^s = \frac{3}{4}I$).

9.3. A structural fact exposed by the test

The repository's entanglement-entropy / central-charge fitter hard-codes local dimension 2, via `psi.reshape(2**n, 2**(L-n))` in three homologous copies. Feeding it the $d = 3$ AKLT state raises `ValueError` immediately.

This is a property of the construction, not a bug to be fixed. It records honestly that this fitter is not applicable to non-spin-1/2 local dimension. The AKLT-side readout therefore takes a d-independent MPS contraction path ($L = 16$, peak memory 0.40 MiB, far below the 16 GiB ceiling; cross-checked bit-for-bit against dense SVD).

9.4. Honest boundaries

1. **This does not show that the ledger is generically applicable to non-critical systems.** It establishes only that *one* criterion is correct on the AKLT chain and that *three* negative controls still fail there. The claims of seven-layer coverage and reusability lie outside the scope of this test.
2. **The AKLT model is textbook material;** it contributes no novelty in physics. What is tested is the framework's behaviour on a non-critical system, not the AKLT chain.
3. **"Bond-dimension reduction" is structurally unrealisable on the open chain**: the literal construction (retaining only one tensor) always yields the zero vector, for which entropy is undefined. It is therefore recorded only on periodic chains, with an added validity pre-condition ($\|\psi\|^2 > 0$) that prevents the zero vector from being passed silently. The open-chain entry is registered as "no object", *not* as "criterion fails".
4. **The three numbers** 78/27/13 **are untouched**, as is the maturity vector (still 1A + 3B + 3C + 0D). The AKLT tests do not enter the negative-control table.

The conclusion is therefore narrow but substantive: the ledger's discriminating power is *not* an artefact of sitting on a critical CFT point. One criterion and three negative controls survive translation to a gapped, area-law, non-CFT system — and one structural limitation of the fitter is exposed in the process.

10. An audit of two cited tensor-network constructions

The ledger audits *our* code. This section asks a different question: do the *cited* tensor-network constructions, on which the upper layers are supposed to sit, hold up at the source level? Both constructions are examined from source snapshots archived in the repository (`v16/Hyper.tex`, `v16/TNR.tex`), so the findings are reproducible independently of our implementation.

10.1. The hyperinvariant MERA (hyMERA) appendix

The construction.. The hyperinvariant (hypersparse) MERA is taken from the cited source [8], archived permanently attributed to `arXiv:1704.04229v1` (`Hyper.tex`). Its Appendix B gives closed-form matrices for an isometry Y and for R, Q, and five reflection matrices, together with multi-tensor constraints $w = A \cdot B \cdot B$ and $u = A^3 B^5$.

What was found.. The core matrix Y fails unitarity outright. The element $(YY^\dagger)_{14}$ is identically $2c_1 s_1 \neq 0$; the bi-unitary residual is the constant $2|c_1 s_1|$, measured at 6.742879×10^{-1}. By raw-diff comparison the transcript is *verbatim* with the source — the defect is in the cited text, not in our transcription. In the same appendix, R, Q, and all five reflection matrices hold exactly (residuals 0 and 2.220446×10^{-16}). The failure is therefore local to Y, not a global typesetting error. Under the multi-tensor constraints, B (which does not contain Y) is exactly unitary, while w and u **both fail**.

A reusable structural conclusion.. Contraction sees only the (12‖34) bipartition: (12‖34) is necessary, whereas (13‖24) is *redundant* among the constraints as written. Independent witness for the latter is supplied by a separate published work [9], which prints a Y that is (12‖34)-unitary (residual 0) but fails the cited source's multi-tensor constraints at (13‖24) with residual 6.743×10^{-1} — *and* satisfies those constraints. The genuine admissibility criterion is therefore:

$$Y \text{ unitary on } (12\|34) \iff \text{multi-tensor constraints hold} \iff \text{the network is a genuine hyMERA.}$$

A positive by-product, and two gaps honestly recorded.. The $\{7, 3\}$ hyperbolic-tiling geometric layer of the construction is internally consistent: a finite patch with $V = 2240$, $E = 2856$, $F = 617$ passes all six combinatorial checks (the algebraic identity at 4.918×10^{-16}, $V - E + F = 1$, connectedness, edge-gluing with no face shared by ≥ 3 tiles, and all interior vertices of valence 3). Assembly of layer V requires only (12‖34) — closing the residual risk of whether layer assembly needs (13‖24). Two gaps are recorded without smoothing: layer V is an abstract cell string, not concentric layers extracted from the tiling patch (the extraction rule is not printed in the source); and the full-layer contraction is not performed.

Why the geometric layer does not connect to our pipeline.. To bridge this network's geometric layer to our c readout would require reshaping the state as $(2^n, 2^{L-n})$, which hard-codes local dimension 2. The legs of this network carry local dimension $\chi = 16$ (A-tensor shape $(16, 16, 16)$). The two premises are therefore mutually exclusive by construction. This is a *criterion that does not hold*, not a numerical regression; following a pre-declared agreement, the attempt was made exactly once, and the negative outcome registered without further investment.

10.2. The tensor-network renormalization (TNR) appendix

The construction.. Tensor-network renormalization (Evenbly and Vidal, 2015) [10] is examined from the archived source snapshot `v16/TNR.tex`, Appendix B.

What was found.. Two equations are mutually exclusive. The right-hand side of Eq. (TRG2) is a Gram sum of real vectors and is therefore *necessarily positive-semidefinite*; yet all 24 of 24 leg orderings of the tensor are non-positive-semidefinite, with minimum eigenvalue -2.0. Hence Eq. (TRG2) cannot hold for *any* real S, independently of index ordering or pairing convention. Of the 12/384 readings under which Eq. (TRG4) can be made to match, *none* contains the printed S. The defect is again at the level of the cited source, not of our contraction convention.

An infrastructural negative result.. A further literature claim — that an existing TNR implementation is available and that porting it is straightforward — was checked against `quimb 1.15.0`. The library exposes no TNR entry point; it has HOTRG and CTMRG, neither of which is TNR. The claim is therefore unanchored.

10.3. Summary of the source-level audit

What this is, and is not. These findings concern *cited* literature, not our own implementation. They are reported because the main ledger's two upper-layer literature interfaces

Construction	Property tested	Outcome
hyMERA, matrix Y	unitarity	**fails**; residual $2\|c_1 s_1\|$ (6.742879×10^{-1}); R, Q, reflections exact
hyMERA, (13‖24) redundancy	necessity of the (13‖24) constraint	**redundant**; witness [9]
hyMERA, $\{7, 3\}$ tiling	combinatorial self-consistency	**passes**; $V = 2240$, $E = 2856$, $F = 617$
hyMERA $\leftrightarrow$ L4 bridge	local-dimension compatibility	**structurally incompatible** ($d = 2$ vs. $\chi = 16$)
TNR, Eq. (TRG2)	positive-semidefiniteness	**fails**; min. eigenvalue -2.0 over 24/24 leg orderings
TNR, Eq. (TRG2) vs. (TRG4)	mutual consistency	**mutually exclusive**
TNR in `quimb 1.15.0`	availability of a TNR entry point	**none**; HOTRG and CTMRG present, neither is TNR

Table 8: Source-level audit of two cited tensor-network constructions. All seven outcomes are reproducible from the archived source snapshots; none changes a reading in the main ledger.

both terminate at the source itself: the hyMERA construction is non-unitary at the source, and the TNR appendix contains mutually exclusive equations. This is a negative result that must sit in the most visible place, not a claim that the constructions cannot be repaired.

11. Updated per-layer honest boundaries (v16.4)

We state these explicitly, in the spirit of the ledger itself. The maturity vector remains A × 1, B × 3, C × 3, D × 0 across all seven layers. **No layer was promoted.**

1. **L1 (B) — The overlap calibrates sensitivity, not independent falsification.** The equal-state overlap rises monotonically ($0.763404 \to 0.938351 \to 0.984428 \to 0.996097 \to 0.999024$) and exceeds 0.99 at τ^*; `dim_full` is consumed for the first time ($65536 = 2^{16}$); rotating the local state drops the overlap below 0.9. This closed the $6/7 \to 7/7$ coverage gap. But the overlap is computed directly from the rotated state, calibrating a sensitivity scale rather than supplying independent physical falsification. And "`equal_state` is the $h \to \infty$ limit" is a mathematical fact; calling it "the L1→L2 causal connection" is a terminology choice — these must not be conflated, since h is finite ($\text{overlap}_\infty = 0.999024 < 1$). L1 remains B.
2. **L2 (A) — Consistency on a finite window, not assumption-free measurement.** The ground-state energy density follows the parameter-free finite-size CFT form. The spectral ratio deviates by 0.212% at $L \leq 16$, but $|c - 0.5|$ grows monotonically from L=10 (5.3×10^{-4}) to L=20 (1.806×10^{-3}, 27.8×), never reversing — falsifying the $O(1/L)$ finite-size-correction interpretation. The ratio drifts *away* from 8 from L=10 onward. Both estimators contain c as a prior; this is self-consistency on a bounded window, not extrapolation to $c = 0.5$. It is the project's single A rating and must be defended against

downgrade, not promoted further without independent corroboration.

3. **L3 (B) — The problem was dissolved, not solved.** The rank-one-ification question was expanded from a single point (N=17) to 7 values of $N \times 5$ seeds $\times$ 8000 steps. Variant B's rank-one-ification is independent of N (5/5 at all 7 N, criterion threshold hit at every point); variant A is 0/5 for $N \leq 17$, 1/5 at N=48, 5/5 at N=64. Since stage 7 follows branch A (A collapses to vacuum), the "N-dependence" question belongs to A — and A does not rank-one-ify at production-accessible N (L$\leq$16 yields only $\{11, 12, 15, 16, 17\}$). A's rank-one direction is a finite-step transient (τ, median 0.4876 vs prediction 1/2); asymptotically it returns to vacuum, not a fixed point. The exponential form matches but the prefactor differs by $\sim$5.9$\times$ (unreproduced). The semantic interpretation of variant B remains open. L3 remains B.
4. **L4 (C) — Descriptive, not implementational.** The MERA entanglement network in L4 is a structural analogy to a discrete geometry; it is not a numerical implementation of AdS/CFT. On the declared grid, MERA curvature measures fall at the 33rd percentile of null-model distributions — i.e., they exhibit negative-curvature features but are not statistical outliers among same-scale random graphs. The Forman curvature used is the triangle-free variant, applicable because the MERA graph has triangle count zero (tree structure); this limits the geometric interpretation. The isometry-disturbance test shows a four-order-of-magnitude gap between the check and its downstream — strong evidence of sensitivity, not holographic emergence. The proposed hyMERA bridge is structurally incompatible with our fitter (local dimension 2 versus $\chi = 16$) and, independently, the cited hyMERA appendix fails unitarity at its core matrix Y (Sec. 10.1). L4 remains C.
5. **L5 (C) — The bridge remains empty.** A controlled recalculation (same parameters, same units, N=36/72/144) eliminated a 252$\times$ unit-confounding artefact but placed the model side inside the reference side's own initial-value spread — a null result. The L4$\rightarrow$L5 bridge therefore remains empty; the 6-dimensional structural comparison claim reduces to 4 effective dimensions (two descriptors degenerate to constants: ϕ collapses to 0.5, k_{p1} is identically the number of wavelengths in-box). The cited TNR appendix contains mutually exclusive equations (Sec. 10.2). L5 remains C.
6. **L6 (B) — "Life" is a distant analogy.** The Gray–Scott system produces Turing patterns; this is a standard reaction–diffusion result, not a model of life origin. The dual-path convergence (10^{-14}–10^{-17}) verifies code fidelity (the equations are solved correctly), not that the equations describe life. Key gradient parameters remain hand-set, not derived from upstream layers. Elevating this to a scientific claim about life would require a dissipative-adaptation or chemical-Darwinism framework, which is outside this work's scope. L6 remains B.
7. **L7 (C) — Reported without a guard; a philosophical endpoint.** A Jacobian spectral radius is reported ($\rho_B = 0.9412 \neq 1$, non-degenerate, with discriminative power) but carries no guard; the log records explicitly that it is *not* pretended to be a negative control. The analytic expectation for branch B was independently derived and verified at 12/12 points (closed-form $\rho_B = g\,(1 - \mathrm{IPR}(x^*))$), but the result is *parameter-dependent*: $\rho_B > 1$ at g = 1.2/1.5 (1.1294/1.4118), so the branch B fixed point *loses stability* beyond $g = 1$, rather than collapsing back to 1.0. This is the expected analytic behaviour for the chosen parameter regime, not an emergent physical result. L7 is the philosophi-

cal/conceptual endpoint of the framework and does not compete with existing scientific theories for verification. The D→C upgrade reflects the addition of a numerical report, not the acquisition of falsifiability. L7 remains C.

Summary of the ladder. The hardest interface is the quantum-entanglement + CFT connection (L2–L3); the softest is gravity/spacetime (L4–L5). The difficulty distribution reflects the actual state of physics: there is no dispute about the former; there is no established theory of the latter. The framework's maturity vector is a map of those gaps, not a reticence about them.

12. Artefacts and reproducibility

The pipeline is driven by a single runner; all figures and the JSON data artefact are regenerated from scratch, and no pre-existing data file is read. The v16 line's incremental probes (five of them) are deliberately kept *outside* the runner's registry: they are diagnostics, and adding them would move the denominators reported in Table 2. Each carries a version tag and writes its own transcript.

Total elapsed time for the frozen reference run is 1119.7 s on an 8-core machine at 49.9% aggregate CPU utilisation, as printed by the runner itself.

12.1. Version provenance

The ledger *numbers* are the v16.3 R-1 full-pipeline run; the *content* of this paper includes the v16.4 iteration's work. The two are not in conflict: every probe added in the later iteration is registered *outside* the runner and *moves no denominator*, precisely so that an earlier frozen ledger stays comparable across versions. That iteration therefore produced no new full-pipeline log, and the v16.3 numbers remain the current ones.

	v15 (initial)	**current (v16.4)**
exit code	`EXIT=0`	`EXIT=0`
scored guards	54/54 = 100.0%	**57/57 = 100.0%**
metrics at target	18/18	**19/19** (27 registered)
diagnostics (excluded)	20 (17 not passing)	**21 (16 not passing)**
negative-control coverage	6/7 layers, gap `['L1']`	**7/7 layers, gap `[]`**
negative controls	12	**13**
maturity vector	A×1, B×3, C×3, D×1	**A×1, B×3, C×3, D×0**
wall time	1245 s	**1119.7 s**

Table 9: The ledger across iterations. All current numbers are from the frozen full-pipeline run `v16/_runall_v16.3_r1.log`. The metric denominator excludes the one layer that carries no target line.

12.2. Which aggregates are well defined, and which are not

The metric layer is reported *by layer*, and one of the seven layers carries no physical target line at all. The registered breakdown is $L1 : 6/6$ (with 2 diagnostics), $L2 : 13/13$ and $L3 : 0/0$ (with 6 diagnostics), giving 19/19 at target among the 27 registered metrics — the remaining 8 being diagnostics excluded from the ratio. The arithmetic closes: $19 + 8 = 27$.

The figure "19/19" is therefore well defined, and we state it. The figure "27/27" is not: it would fold in eight items that were never given a threshold, and would count a layer whose registered distribution is 0/0 as though it had passed something. Quoting it would be a category error — and one that inflates, which is the direction that matters. Similarly, the metric and guard ledgers must never be summed: they are disjoint and share no denominator.

12.3. Expected output

Line	**Expected**
`metrics meeting target`	`19/19`
`guard pass rate`	`57/57 = 100.00%` (plus 21 diagnostics excluded, 16 of them not passing)
`ledger completeness`	`phantom references: none; missing layers: none`
negative-control layer coverage	7/7, gap `[]`
`MERA bulk geometry`	Forman mean -2.242, Ollivier mean -0.3297
wall clock	~ 1119.7 s (8 cores); order-of-magnitude reference

Table 10: Expected output for a correct reproduction.

Two cautions. *(a)* These numbers certify that the pipeline *ran correctly*. They do not certify that any conclusion is *true*: 57/57 means 57 scored assertions hold. *(b)* The values in Table 10 are transcribed from the frozen reference run. The wall-clock figure is machine-dependent and should be read as an order of magnitude only.

12.4. Code and data availability

Code availability. The complete source tree, frozen at version v16.9, is available under the Apache-2.0 licence at `https://github.com/yanjianzhong/spiral-emergence`. The repository is maintained with all commits DCO-signed (`Signed-off-by`); the v16 release tag and its assets are immutable. The source snapshots used for the source-level audit in Sec. 10 (`v16/Hyper.tex`, `v16/TNR.tex`) are archived within the repository and permanently attributed to `arXiv:1704.04229v1`.

Data availability. The frozen full-pipeline log `v16/_runall_v16.3_r1.log`, all runner transcripts, and the AKLT-chain logs (`_t9_aklt.log` and the two frozen intermediate logs) are included in the repository above. No external dataset was used; all numerical artefacts are regenerated from source. Zenodo concept DOI (all versions): `https://doi.org/10.5281/zenodo.22810161`; v16.9 DOI: `https://doi.org/10.5281/zenodo.23278413`.

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13. Related work

Tensor networks and MERA. The Multi-scale Entanglement Renormalization Ansatz (MERA) was introduced by Vidal [11] and developed into a binary-graph implementation by Evenbly and Vidal [3]. Its application to holographic duality and emergent geometry was motivated by Swingle [2], who noted that entanglement renormalization naturally produces a hyperbolic geometry in the bulk. In homogeneous MERA, many structural properties are predetermined by the network geometry and largely independent of variational parameters [5], which is the context for our negative result in Sec. 5.4. The hyperinvariant MERA variant [8] and tensor-network renormalization [10] are the subjects of the source-level audit in Sec. 10.

The AKLT model. The spin-1 chain with SU(2)-invariant nearest-neighbour projector Hamiltonian was introduced by Affleck, Kennedy, Lieb, and Tasaki [6, 7]. Its exact $\chi = 2$ MPS ground state and Haldane gap make it a standard gapped-area-law benchmark and the substrate of the scope-negative-control in Sec. 9.

Negative controls in experiment. The criterion used here is the ordinary logic of a controlled experiment, transplanted into software. The transplant is the point: in a wet-lab experiment a negative control is a physical sample; in a pipeline it must be an *executable* alternative, and executable alternatives are easy to omit without noticing.

Property-based testing and metamorphic testing. The repulsion and two-path types are close relatives of metamorphic relations and property-based tests, in that they assert a relation that must hold across a transformation rather than a fixed expected output. The difference is one of framing: these controls are attached to *scientific claims about a physical system*, and their failure is meant to propagate into how loudly the claim is stated.

Reproducibility infrastructure. Workflow engines and provenance systems record *what ran*. This work is about a complementary question: *what would have to be true for the result to be wrong*, and whether the pipeline would notice.

Model validation in the physical sciences. Null-model comparison is standard practice in network science, and the null-model control here (Sec. 2.2) is borrowed directly from that tradition — including the discipline of reporting *which* nuisance structure each null destroys.

What traditional sanity checks miss. The most direct way to appreciate the value of the negative-control discipline is to ask: which defects in this pipeline would *not* have been caught by conventional positivity, finiteness, or non-emptiness checks? Four concrete cases answer that question:

1. **An inverted completeness criterion** (Sec. 4.4): a guard that *passed* when the referenced control was missing and *failed* in the healthy case. Every positivity and finiteness check in the pipeline reported green throughout. Only the completeness guard — which checks that cited guard names exist in the runtime registry — could detect this, because the defect was in the *accounting structure*, not in any numerical value.
2. **A fully wired bridge carrying no upstream quantity** (Sec. 5.3): layer 4 derives $\chi_{\text{dev}} = 4$ from the area law, but layer 5 hard-codes its own bond dimension. The call chain is complete end-to-end; every function returns a finite, non-empty result. A positivity check on the curvature report would have passed silently. The gap is visible only to a control that asks whether the *upstream-derived quantity* actually flows to the consumer.

3. **A readout invariant under the fitted state** (Sec. 5.4): the MERA bulk graph is bit-identical across seeds, bond dimensions, and *states*. A sanity check asking "is the graph non-empty?" or "is curvature finite?" passes trivially. Only the explicit invariance test (rebuild and compare edge sets) reveals that the readout is an analytic property of the wiring, not an emergent geometry.
4. **A guard that passed on an overflowed norm** (Sec. 6.5): the finiteness check — a member of the explicitly excluded family — passed on a vector whose norm had silently overflowed; the program printed $S(L/2) = 0$ with no warning. There is no tolerance, step count, or system size at which this becomes visible. It becomes visible only when one asks what the check's increment means.

These are not hypothetical vulnerabilities. They are defects that *existed in the pipeline as built*, and that the negative-control ledger caught precisely because each control is required to be able to fail. A positivity check cannot catch what is already positive.

14. Conclusion

We stated an operational criterion — a negative control must be able to fail when its conclusion is false — and used it as a filter rather than an aspiration, excluding positivity, finiteness, and non-emptiness checks outright. The survivors were sorted into four types; a per-layer ledger was built with a completeness guard that checks against the *runtime* registry; and scored assertions were separated from diagnostics so the pass rate cannot be inflated by checks that cannot fail.

Applied to a seven-layer pipeline whose engine is a transverse-field Ising / MERA tensor-network construction, the method produced coverage of all 7 layers and, more usefully, a series of negative results easy to omit: a layer with a structural gap (nothing to push against); a fully wired bridge carrying no upstream-derived quantity; a "geometry" readout that is an analytic property of the network wiring; and a cMERA terminal state indistinguishable from the bare vacuum.

Turning the discipline on the ledger itself found five cases where *a check incremented correctly while the thing it stood for did not obtain*: coverage increased while independence did not; a derivation was present while no derivation occurred; a registry entry existed while its described behaviour did not; a correction was applied while the refuted claim survived elsewhere; and a finiteness guard — written by an author who had excluded finiteness checks — passed on a vector whose norm had silently overflowed, printing $S(L/2) = 0$ with no warning. There is no tolerance, step count, or system size at which that last case becomes visible; it becomes visible only when one asks what the check's increment *means*.

The scope of the ledger was then tested in two directions. Feeding the exact $\chi = 2$ MPS ground state of the gapped, non-CFT AKLT chain to the gapped-state criterion returns the known area law ($c = 0$); three constructive negative controls continue to fail as designed, and the local-dimension limitation of the c fitter is exposed. An audit of two cited tensor-network constructions finds source-level defects: the core matrix Y of the hyMERA appendix fails unitarity with residual $2|c_1 s_1|$, while the TNR appendix contains a necessarily positive-semidefinite Gram form whose tensor is non-positive-definite in all 24 leg orderings, and two of its equations are mutually exclusive. These findings are about the cited literature, not our

implementation; they are reported as negative results, and close the question of where the upper-layer interfaces terminate.

Self-falsifiability. The ledger's completeness guard fails if a cited guard does not exist in the runtime registry or if a layer is missing. These are checks on the accounting, which is precisely their scope: they make the ledger *hard to fake*, not the physics true. This is the method's most distinctive property: **the validation framework is itself validated by the same standard it enforces**. The fact that this self-check caught a real defect — an inverted completeness criterion that passed for the wrong reason — is evidence that the discipline works *on itself*.

What this paper is not. It is not a theory of everything; it does not establish the seven-layer framework; it **not** claim that MERA describes physical systems. The contribution is methodological: this paper validates **the feasibility and limitations of a self-auditing procedure**, using a tensor-network pipeline as a *computational testbed*. It claims no physical result beyond textbook Ising criticality. Its value lies in the discipline of accounting for what one cannot check, and in showing that this discipline, aimed at itself, finds things worth reporting.

Appendix A. Guard accounting, verbatim

Log line, quoted without editing:

```
guards 57/57 = 100.00% (plus 21 diagnostics excluded, of which 16 did not pass)
```

Diagnostic registration is explicit per guard (`expect_pass=False`); 78 guards total = 57 scored + 21 diagnostic. Of the 21 diagnostics, 16 fail by design and 5 pass while remaining excluded from the denominator. The distinction between "failed" and "defective" is therefore carried by the data structure and not by the reader's charity.

A methodological note, recorded deliberately. The suspicion that the bond dimension constituted an undeclared coupling between layers 4 and 5 — described in Sec. 5.3 — was raised and then falsified within seconds, because the cheapest available falsification test was named *before* any effort was spent defending the claim. This was the third time in the project that a causal claim was overturned by its own measurement. The practice of stating the cheapest falsification first is worth more than any individual result in this paper, and is recommended independently of it.

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